

Commerce XI Maths App

Theory (Pages 1-9) - Detailed Answers with Explanations

Total questions: 35 | +2 correct | -1 wrong

This answer guide follows the question order used in the app.

Question 1 - Theory Q1

Which collection is a set?

Options

- A. Days in a week
- B. Happy people in a town
- C. Clever students in a class
- D. Successful people in a city

Correct answer: Days in a week

Explanation and steps

1. The governing definition or property gives: Days in a week.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Days in a week.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 2 - Theory Q2

A set is a collection of objects that are:

Options

- A. Well-defined
- B. Similar
- C. Countable
- D. Written in order

Correct answer: Well-defined

Explanation and steps

1. The governing definition or property gives: Well-defined.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Well-defined.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 3 - Theory Q3

If x is an element of A , which notation is correct?

Options

- A. $x \in A$

- B. $x \subseteq A$
- C. $A \in x$
- D. $x \notin A$

Correct answer: $x \in A$

Explanation and steps

1. The symbol $\in$ means "is an element of". Since x is an element of A , we write $x \in A$.
2. The symbol $\subseteq$ compares one set with another set, so $x \subseteq A$ is not the required element notation.
3. $A \in x$ reverses the stated relationship, while $x \notin A$ means that x is not an element of A .
4. Therefore, the correct answer is $x \in A$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 4 - Theory Q4

In the Roster method, elements are:

Options

- A. Listed inside braces and separated by commas
- B. Shown only in a diagram
- C. Described by a property
- D. Listed without brackets

Correct answer: Listed inside braces and separated by commas

Explanation and steps

1. The governing definition or property gives: Listed inside braces and separated by commas.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Listed inside braces and separated by commas.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 5 - Theory Q5

What is the set of letters in ABBA in Roster form?

Options

- A. $\{A, B\}$
- B. $\{A, B, B, A\}$
- C. $\{A, A, B, B\}$

D. {AB, BA}

Correct answer: {A, B}

Explanation and steps

1. The governing definition or property gives: {A, B}.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is {A, B}.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 6 - Theory Q6

Does changing the order of elements change a set?

Options

- A. No
- B. Yes
- C. Only for finite sets
- D. Only for number sets

Correct answer: No

Explanation and steps

1. The governing definition or property gives: No.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is No.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 7 - Theory Q7

The Set-Builder method describes elements by:

Options

- A. A property that determines them
- B. Their order
- C. A diagram
- D. Their number alone

Correct answer: A property that determines them

Explanation and steps

1. The governing definition or property gives: A property that determines them.

2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is A property that determines them.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 8 - Theory Q8

A Venn diagram is a:

Options

- A. Pictorial representation of a set
- B. List of every element
- C. Formula for cardinality
- D. Type of interval

Correct answer: Pictorial representation of a set

Explanation and steps

1. The governing definition or property gives: Pictorial representation of a set.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Pictorial representation of a set.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 9 - Theory Q9

Which interval excludes both a and b?

Options

- A. (a, b)
- B. $[a, b]$
- C. $[a, b)$
- D. $(a, b]$

Correct answer: (a, b)

Explanation and steps

1. The governing definition or property gives: (a, b) .
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is (a, b) .

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 10 - Theory Q10

Which interval includes both a and b?

Options

- A. $[a, b]$
- B. (a, b)
- C. $[a, b)$
- D. $(a, b]$

Correct answer: $[a, b]$

Explanation and steps

1. The governing definition or property gives: $[a, b]$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $[a, b]$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 11 - Theory Q11

Which interval includes a but excludes b?

Options

- A. $[a, b)$
- B. (a, b)
- C. $(a, b]$
- D. $[a, b]$

Correct answer: $[a, b)$

Explanation and steps

1. The governing definition or property gives: $[a, b)$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $[a, b)$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 12 - Theory Q12

Which interval excludes a but includes b?

Options

- A. $(a, b]$

- B. $[a, b)$
- C. $[a, b]$
- D. (a, b)

Correct answer: $(a, b]$

Explanation and steps

1. The governing definition or property gives: $(a, b]$.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is $(a, b]$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 13 - Theory Q13

$n(A)$ means the number of:

Options

- A. Distinct elements in A
- B. Sets equal to A
- C. Subsets of A
- D. Elements outside A

Correct answer: Distinct elements in A

Explanation and steps

1. The governing definition or property gives: Distinct elements in A.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Distinct elements in A.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 14 - Theory Q14

A set with no element is called a:

Options

- A. Empty set
- B. Singleton set
- C. Universal set
- D. Power set

Correct answer: Empty set

Explanation and steps

1. The governing definition or property gives: Empty set.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Empty set.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 15 - Theory Q15

A set containing exactly one element is a:

Options

- A. Singleton set
- B. Finite set only
- C. Universal set
- D. Proper subset

Correct answer: Singleton set

Explanation and steps

1. The governing definition or property gives: Singleton set.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Singleton set.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 16 - Theory Q16

Is the empty set finite?

Options

- A. Yes
- B. No
- C. Only when the universal set is finite
- D. Only when it is written as $\{ \}$

Correct answer: Yes

Explanation and steps

1. The governing definition or property gives: Yes.
2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is Yes.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 17 - Theory Q17

Which is an infinite set?

Options

- A. Natural numbers
- B. Vowels in English
- C. Days in a week
- D. Letters in "beautiful"

Correct answer: Natural numbers

Explanation and steps

1. The governing definition or property gives: Natural numbers.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Natural numbers.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 18 - Theory Q18

Two sets are equal when they have:

Options

- A. The same elements
- B. The same number of elements
- C. The same order of elements
- D. The same name

Correct answer: The same elements

Explanation and steps

1. The governing definition or property gives: The same elements.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is The same elements.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 19 - Theory Q19

Two finite sets are equivalent when they have:

Options

- A. The same number of elements
- B. Exactly the same elements
- C. No common elements
- D. The same power set

Correct answer: The same number of elements

Explanation and steps

1. The governing definition or property gives: The same number of elements.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is The same number of elements.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 20 - Theory Q20

$A \subseteq B$ means:

Options

- A. Every element of A is in B
- B. Every element of B is in A
- C. A and B have equal sizes
- D. A is empty

Correct answer: Every element of A is in B

Explanation and steps

1. The governing definition or property gives: Every element of A is in B.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Every element of A is in B.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 21 - Theory Q21

If A is a proper subset of B, then:

Options

- A. Every element of A is in B, and B has at least one element not in A
- B. $A = B$

- C. A has more elements than B
- D. A and B are disjoint

Correct answer: Every element of A is in B, and B has at least one element not in A

Explanation and steps

1. The governing definition or property gives: Every element of A is in B, and B has at least one element not in A.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Every element of A is in B, and B has at least one element not in A.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 22 - Theory Q22

A universal set for a discussion contains:

Options

- A. Every set under consideration as a subset
- B. Only natural numbers
- C. Only one element
- D. No elements

Correct answer: Every set under consideration as a subset

Explanation and steps

1. The governing definition or property gives: Every set under consideration as a subset.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Every set under consideration as a subset.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 23 - Theory Q23

The power set $P(A)$ contains:

Options

- A. All subsets of A
- B. Only the elements of A
- C. Only proper subsets of A
- D. Only the empty set

Correct answer: All subsets of A

Explanation and steps

1. The governing definition or property gives: All subsets of A.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is All subsets of A.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 24 - Theory Q24

Relative to universal set U, A' contains:

Options

- A. Elements in U that are not in A
- B. All elements of A
- C. Elements common to A and U
- D. Elements outside U

Correct answer: Elements in U that are not in A

Explanation and steps

1. The governing definition or property gives: Elements in U that are not in A.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Elements in U that are not in A.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 25 - Theory Q25

Which identity is correct?

Options

- A. $(A')' = A$
- B. $(A')' = \emptyset$
- C. $(A')' = U$
- D. $(A')' = P(A)$

Correct answer: $(A')' = A$

Explanation and steps

1. The governing definition or property gives: $(A')' = A$.
2. Compare each option with that definition or with the calculation required by the question.

3. Therefore, the correct option is $(A')' = A$.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 26 - Theory Q26

$A \cup B$ contains elements in:

Options

- A. A or B, including those in both
- B. Both A and B only
- C. A but not B
- D. Neither set

Correct answer: A or B, including those in both

Explanation and steps

1. The governing definition or property gives: A or B, including those in both.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is A or B, including those in both.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 27 - Theory Q27

$A \cap B$ contains elements in:

Options

- A. Both A and B
- B. A or B
- C. A but not B
- D. Neither set

Correct answer: Both A and B

Explanation and steps

1. The governing definition or property gives: Both A and B.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Both A and B.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 28 - Theory Q28

If $A \cap B = \emptyset$, the sets are:

Options

- A. Disjoint
- B. Equal
- C. Equivalent
- D. Universal

Correct answer: Disjoint

Explanation and steps

1. The governing definition or property gives: Disjoint.
2. Compare each option with that definition or with the calculation required by the question.
3. Therefore, the correct option is Disjoint.

Option check: the remaining choices do not satisfy the definition, calculation, membership condition, endpoint rule, or ordered-pair direction used above.

Question 29 - Theory Q29

If $A = \{a, b, a, c\}$, enter $n(A)$.

Correct answer: 3

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 3.
3. Therefore, the required answer is 3.

Question 30 - Theory Q30

If $A = \{p, q, r, s\}$, enter $n(P(A))$.

Correct answer: 16

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 16.
3. Therefore, the required answer is 16.

Question 31 - Theory Q31

If $n(A) = 12$, $n(B) = 9$, and $n(A \cap B) = 4$, enter $n(A \cup B)$.

Correct answer: 17

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 17.
3. Therefore, the required answer is 17.

Question 32 - Theory Q32

If $n(U) = 25$ and $n(A) = 8$, enter $n(A')$.

Correct answer: 17

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 17.
3. Therefore, the required answer is 17.

Question 33 - Theory Q33

If $n(A \cup B) = 18$, $n(A) = 10$, and $n(B) = 12$, enter $n(A \cap B)$.

Correct answer: 4

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 4.
3. Therefore, the required answer is 4.

Question 34 - Theory Q34

If $n(A) = 10$ and $n(A \cap B) = 3$, enter $n(A - B)$.

Correct answer: 7

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 7.
3. Therefore, the required answer is 7.

Question 35 - Theory Q35

If $n(A) = 12$, $n(B) = 10$, $n(C) = 8$, $n(A \cap B) = 5$, $n(B \cap C) = 4$, $n(A \cap C) = 3$, and $n(A \cap B \cap C) = 2$, enter $n(A \cup B \cup C)$.

Correct answer: 20

Explanation and steps

1. Use the definition, identity, or counting formula stated in the question.
2. Substituting the given values and simplifying gives 20.
3. Therefore, the required answer is 20.